# Abderrahmene Boudiaf

Computer Vision · Foundation Models · AI for Agriculture

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## Research Summary

PhD candidate in the Department of Electrical Engineering and Computer Science at **Khalifa University of Science and Technology** (Abu Dhabi, UAE), working within the **Visual Recognition Group** under the supervision of Dr. Sajid Javed. My research sits at the intersection of **computer vision**, **vision–language models (VLMs)**, and **multimodal large language models (MLLMs)**, with a primary application domain in **precision agriculture and crop analysis**. I design foundation-model pipelines—domain-adapted VLMs, retrieval-augmented segmentation, and multimodal chatbots—that translate state-of-the-art AI into scalable tools for sustainable food production.

**Research Interests:** Vision–Language Models · Foundation Models · Semantic Segmentation · Few-Shot Learning · Multimodal LLMs · AI for Agriculture · Open-Vocabulary Recognition · Retrieval-Augmented Generation · Underwater Vision

## Education

**PhD, Electrical Engineering & Computer Science** *2020 – Present (Expected December 2026)*

Khalifa University of Science and Technology, Abu Dhabi, UAE
Thesis: *Computer Vision for Agriculture and Crops*
Supervisor: Dr. Sajid Javed (Computer Vision Group)

**MSc, Robotics & Autonomous Systems** *2020 – 2022*

Khalifa University of Science and Technology, Abu Dhabi, UAE
Thesis: *Robot Navigation in Low Visibility Conditions*
Supervisor: Prof. Jorge Dias (Robotics Group)

## Research Metrics (Google Scholar, May 2026)

| Total Publications | Total Citations | h-index | i10-index |
|---|---|---|---|
| 12 | 55 | 4 | 2 |

## Publications

### Agriculture & Foundation Models for Vision

**CropVLM: A Domain-Adapted Vision-Language Model for Open-Set Crop Analysis**
A. Boudiaf, S. Javed
*arXiv:2605.03259 (2026)* *Code: github.com/boudiafA/CropVLM*
VLM adapted via Domain-Specific Semantic Alignment on 52,987 image–caption pairs spanning 37 crop species in natural field conditions. Introduces HOS-Net enabling zero-shot crop detection from text descriptions only: **72.51% zero-shot accuracy**, 49.17 $AP_{50}$ on novel tropical fruit species—no retraining required.

**AgriChat: A Multimodal Large Language Model for Agriculture Image Understanding**
A. Boudiaf, I. Hussain, S. Javed
*arXiv:2603.16934 (2026)* *Code: github.com/boudiafA/AgriChat*

LoRA fine-tuned MLLM on AgriMM: >121k images, >607k QA pairs, 3,000+ plant classes. V2VK pipeline eliminates biological hallucinations by grounding training data in verified phytopathological literature. Supports plant species identification, disease diagnosis, and fruit counting.

## Retrieval-Augmented & Open-Vocabulary Segmentation

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### SegRAG: Training-Free Retrieval-Augmented Semantic Segmentation

A. Boudiaf, I. Hussain, S. Javed

*arXiv:2605.17630 (2026)*

Training-free framework that augments open-vocabulary segmentation with RAG, enabling dense prediction on arbitrary categories without model retraining.

## CLIP, Few-Shot Learning & Anomaly Detection

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### CLIFS: CLIP-Driven Few-Shot Learning for Baggage Threat Classification

A. Ahmed, D. Velayudhan, M. ElMezain, M. Al Radi, A. Boudiaf, T. Hassan, *et al.*

*IEEE International Conference on Image Processing (ICIP 2024), pp. 753–759.* ***[11 citations]***

### SMO-CLIP: Enhancing Anomalous Smoke Density Assessment Using a Hybrid LLM–VLM Approach

P. Li, M. Al Radi, M. S. Elmezain, A. H. Ahmed, A. Boudiaf, S. Boumaraf, J. Dias, *et al.*

*IEEE International Conference on Image Processing (ICIP 2024), pp. 760–765.* ***[4 citations]***

## Image Enhancement & Underwater Computer Vision

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### Underwater Image Enhancement Using Pre-trained Transformer

A. Boudiaf, Y. Guo, A. Ghimire, N. Werghi, G. De Masi, S. Javed, J. Dias

*ICIAP 2022, Lecture Notes in Computer Science vol. 13233, pp. 480–488, Springer.* ***[29 citations — highest cited work]***

Pre-trained denoising transformers applied to underwater image restoration, achieving state-of-the-art quality improvement without domain-specific retraining. Funded by KU awards RC1-2018-KUCARS and CIRA-2019-047.

### BB-UWCrack: A Benchmark Dataset for Bounding Box Crack Detection in Underwater Structures

M. ElMezain, A. Boudiaf, M. Abujabal, A. Ghimire, G. De Masi

*Advances in Science & Engineering Technology International Conferences (ASET 2025)*

## Robotics & Autonomous Systems

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### Image-Based Obstacle Avoidance Using 3DConv Network for Rocky Environments

A. Boudiaf, A. Al Sumaiti, J. Dias

*IEEE International Symposium on Robotic and Sensors Environments (ROSE 2022).* ***[5 citations]***

### Reinforcement Learning-Based Optimal Control Design for Active Suspension Systems

A. Ahmad, M. Al Radi, A. Ahmad, A. Boudiaf

*Advances in Science & Engineering Technology International Conferences (ASET 2024).* ***[5 citations]***

### Robotic Optimal Path Planning via Multi-Objective Dolphin Echolocation Optimization (DEO)

M. Al Radi, A. Ahmad, A. Boudiaf

*Advances in Science & Engineering Technology International Conferences (ASET 2025)*

### Biomedical Signal Analysis

**Machine Learning-Based Multivariate Classification of Cardiac Autonomic Neuropathy via Imbalanced ECG Recordings**
M. Al Radi, A. Ahmad, A. Ahmad, A. Boudiaf, N. Alkaabi, M. Malouf, H. Jelinek
*Advances in Science & Engineering Technology International Conferences (ASET 2024).* ***[1 citation]***

### Self-Supervised Learning & Marine Vision

**Self-Supervised Plankton Classification via DINO and Gradient-Based Loss Re-weighting**
A. Ahmad, Z. Habash, A. Ahmad, M. Alradi, A. Boudiaf, N. Werghi, M. Owais, *et al.*
*[Under Review], 2026*

### MSc Thesis

**Robot Navigation in Low Visibility Conditions**
A. Boudiaf
*Khalifa University of Science and Technology, Abu Dhabi, UAE, 2022*

## Research Experience

**PhD Researcher — Visual Recognition Group, Khalifa University** *2020 – Present*

- Designed and open-sourced **CropVLM** and **AgriChat**: foundation-model pipelines for agricultural AI covering 3,000+ crop classes and 37 species.
- Built **SegRAG**: a training-free retrieval-augmented semantic segmentation framework using open-vocabulary VLMs.
- Led underwater image enhancement work using pre-trained transformers (29 citations, ICIAP 2022, Springer LNCS).
- Co-authored two IEEE ICIP 2024 papers on CLIP-based few-shot threat classification and hybrid LLM–VLM anomaly detection.
- Contributed to three externally funded KU projects (RC1-2018-KUCARS, CIRA-2019-047, FSU-2022-003).
- Mentored junior MSc and PhD researchers; supported grant deliverables and technical reporting.

**MSc Researcher — Khalifa University** *2020 – 2022*

- Developed a 3D convolutional network for image-based obstacle avoidance in rocky terrain (IEEE ROSE 2022, 5 citations).
- Investigated autonomous robot navigation under degraded visual conditions (fog, dust, low light).

## Teaching Experience

**Teaching Assistant — Khalifa University of Science and Technology** *2020 – Present*

**Data Science and Artificial Intelligence** Lab Instructor
Khalifa University of Science and Technology, Abu Dhabi, UAE

- Conducted hands-on lab sessions covering data analysis fundamentals, statistical methods, and machine learning concepts.

- Instructed students in Python programming for AI applications, including data manipulation with NumPy and Pandas, visualisation, and model development with scikit-learn.
- Guided students through practical assignments and projects applying AI techniques to real-world datasets.

**Introduction to MATLAB** Lab Instructor
Khalifa University of Science and Technology, Abu Dhabi, UAE

- Taught foundational MATLAB programming concepts including scripting, matrix operations, and data visualisation.
- Designed and delivered practical lab exercises to reinforce programming skills and engineering problem-solving using MATLAB.

## Technical Skills

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|---|---|
| **Deep Learning & CV** | PyTorch, JAX, HuggingFace Transformers, CLIP, LLaVA, SAM, ViT, DINO, Deformable DETR, OpenCLIP |
| **LLM / MLLM** | LoRA fine-tuning, RAG pipelines, Instruction tuning, VL alignment, Prompt engineering |
| **Agriculture AI** | Plant phenotyping, Disease classification, Crop detection, Open-set recognition, Agronomic VQA |
| **Programming** | Python, MATLAB, C++ (basic) |
| **Tools & Platforms** | AWS, Docker, Git, Weights & Biases, OpenCV, scikit-learn, NumPy, Pandas |
| **Robotics** | ROS, Path planning, Reinforcement learning, Optimal control |
| **Languages** | Arabic (native), English (fluent) |

## References

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